

NGUYEN LUONG

Data Scientist & Machine Learning Engineer

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Data scientist and ML engineer with 4+ years of research experience and 4+ years in production software development. Currently a Research Software Engineer at Aalto University with a focus on MLOps practice. PhD candidate specializing in computational social science, with deep expertise in statistical modeling and multi-modal sensor data fusion.

TECHNICAL SKILLS

ML / Data Science	Statistical modeling, Deep learning, Time-series analysis
Frameworks	PyTorch, PyTorch Lightning, scikit-learn, XGBoost, LightGBM, Optuna, Weights & Biases
Languages	Python, R, SQL, Swift
Data Engineering	Pandas, Polars, DuckDB, Apache Spark (PySpark)
Cloud & MLOps	Docker, Git, CI/CD, MLflow
Visualization	Matplotlib, Seaborn, Plotly, Vega & Altair

EXPERIENCE

Research Software Engineer | Aalto University, Finland

Nov 2025 – Present

Department of Computer Science

- ▶ Consulting researchers across multiple groups on computational methods: experimental design, model selection, pipeline architecture
- ▶ Teaching foundational and advanced computational skills to researchers, topics include: Python best practices, version control (Git), containerization (Docker), HPC/Slurm workflows, and reproducible research
- ▶ Designing and implementing MLOps infrastructure: experiment tracking (MLflow, W&B), optimization (Tensorboard, Optuna), model versioning, automated training pipelines
- ▶ Building CI/CD pipelines for research codebase: automated testing, linting, documentation generation, and package publishing to PyPI

PhD Candidate | Aalto University, Finland

Nov 2021 – Present

Advisor: Dr. Talayeh Aledavood

- ▶ Built statistical and ML models (Python, R) analyzing daily behavioral patterns from smartphone and wearable sensor data across 500+ participants
- ▶ Designed longitudinal studies analyzing COVID-19 impact on circadian rhythms using multi-sensor fusion (accelerometer, GPS, screen time, sleep staging)
- ▶ Published 8 peer-reviewed papers in top venues: JAMIA, JMIR, and SoftwareX

Senior iOS Developer | ParkMan, Finland

Dec 2016 – Dec 2020

Leading parking application in Finland — 200k+ monthly active users

- ▶ Led iOS development: managed product roadmap, architecture decisions, and App Store releases for a consumer-facing fintech application

- ▶ Reduced churn rate by 10% by redesigning user journey using a data-driven A/B testing approach with analytics-informed decision making
- ▶ Doubled deployment cycle (monthly → bi-weekly) by building a CI/CD pipeline with automated testing and release management
- ▶ Executed large-scale codebase migration from Objective-C to Swift, improving maintainability and reducing crash rate

EDUCATION

MSc in Data Science, Machine Learning, and AI | Aalto University, Finland

Jun 2021

Graduated with honors

- ▶ Thesis: Estimating treatment effects from clinical data using a Transformer-based model (causal inference + deep learning)
- ▶ Advisor: Prof. Pekka Marttinen

BSc in Computer Science | Metropolia AMK, Finland

Jun 2017